



Emergency Lighting Self-contained luminaires



Powering Business Worldwide

A long history of expertise in the life safety industry with a commitment to deliver market-leading solutions that protect people and property.

Heritage of innovation.

CEAG
Cooper
Menvier
JSB
Luminox
Crompton
Blessing
Univel



When it comes to protecting life and property, there's no room to compromise.

In a constantly changing world, owners and operators of commercial and industrial buildings must keep up-to-date with the changing nature of risk. Safe evacuation is becoming more challenging due to a number of external influences.

What are the risks businesses face today ?

The ongoing risk of fire

Over a third of businesses never resume operations after a major fire - losing orders, contracts, and key employees. This results in lost jobs and services to the community.

Non-traditional threats

Power outages, terrorism and domestic extremism are a rising cause for evacuation. These risks demand a different approach when planning for safe evacuation. High-profile terrorist attacks can shape regulation.

Each emergency lighting system is important, it protects life and health.

- Escape route marking during regular power supply: Evacuation of a building due to an accident, a bomb threat etc.
- During blackout: Light supply
 - Showing the directions out of the building
 - Illumination of the escape route to guarantee a safe evacuation



Our products meet your challenges

Innovation, tests and compliance for more reliability

- We constantly innovate for contemporary design and technologies
- Customers light engineering requirements are fully tested at an in-house lighting laboratory. We also expose newly developed products to extreme conditions and life cycle testing
- As a commitment to deliver high quality for all products and employees, Eaton's emergency lighting manufacturing facilities are certified ISO 9001
- Most of our products and complete systems are 3rd party certified

Different technologies to meet the requirements of your building and activity

- "Unmonitored" luminaires: only the battery charge circuit is tested, manual tests and inspection book management
- Luminaire Automatic Test (AT): battery and light source circuit is continuously monitored and the status is displayed via LED. Specified tests (Weekly function test, annual duration test) are automatically started.
- Addressable luminaires: luminaires monitored and automatic tests, central display of system status, automatic inspection book management with digital memory, web access and visualisation possible.

Eco-friendly luminaires all along their life cycle

Our manufacturing plants are ISO 14001 & ISO 9001 certified. We are committed to favour the choice of recycled materials and reduce weight and volume of products and packaging. Our LED luminaires are low consumption, prevent from relamping operations as their life time goes up to 60.000 hours.

A large portfolio for a wide range of applications

- Exit sign, escape route, anti-panic luminaires
- Indoor or with high degree of protection
- Aesthetic solutions
- High output luminaires and beam light projectors for large premises

Look to the future, trust our past

The projects listed below are only a selection of the locations and applications where EATON emergency lighting solutions are installed.



Hospitality & leisure

- New Modern Art Museum, Italy
- Ritz-Carlton Hotel, Germany
- Marriot Opera Ambassador, France
- Ramada Resort Hotel, Hungary
- Atlantis the Palm Hotel, Dubai



Schools and universities

- Technical University Berlin, Germany
- University Hamburg, Germany
- Normale University, Italy
- University Zurich, Switzerland
- University of Dubai, U.A.E.



Airports

- Frankfurt, Germany
- Athens, Greece
- Schiphol, Netherlands
- Bangkok, Thailand
- Dubai, United Arab Emirates



Sport venues

- Emirates Stadium, UK
- Stadium Borussia-Park, Germany
- Stade des Lumières Lyon, France
- Karaiskakis Stadium, Greece
- National Aquatics Center, China



High-rise buildings

- Tower 115, Slovakia
- Etisalat Tower, U.A.E.
- Capital Gate Tower, U.A.E.
- Burj Khalifa Tower, U.A.E.
- Burj Al Arab, U.A.E.



Commercial centres / retail

- Centro, Germany
- Carrefour Supermarkets, Belgium
- Arena Plaza Budapest, Hungary
- Montedoro Freetime, Italy
- Mazaya Centre, U.A.E.



Industry

- Microsoft, Italy
- EADS Airbus, Germany & France
- Bayer, Germany
- BP, Norway
- Dubai Cable Company, Abu Dhabi



Assembly halls / rooms

- German Bundestag, Germany
- National Theatre Budapest, Hungary
- National Library Leipzig, Germany
- Town Hall Sydney, Australia
- National Convention Center, Qatar



Eaton answers any specific application requirements

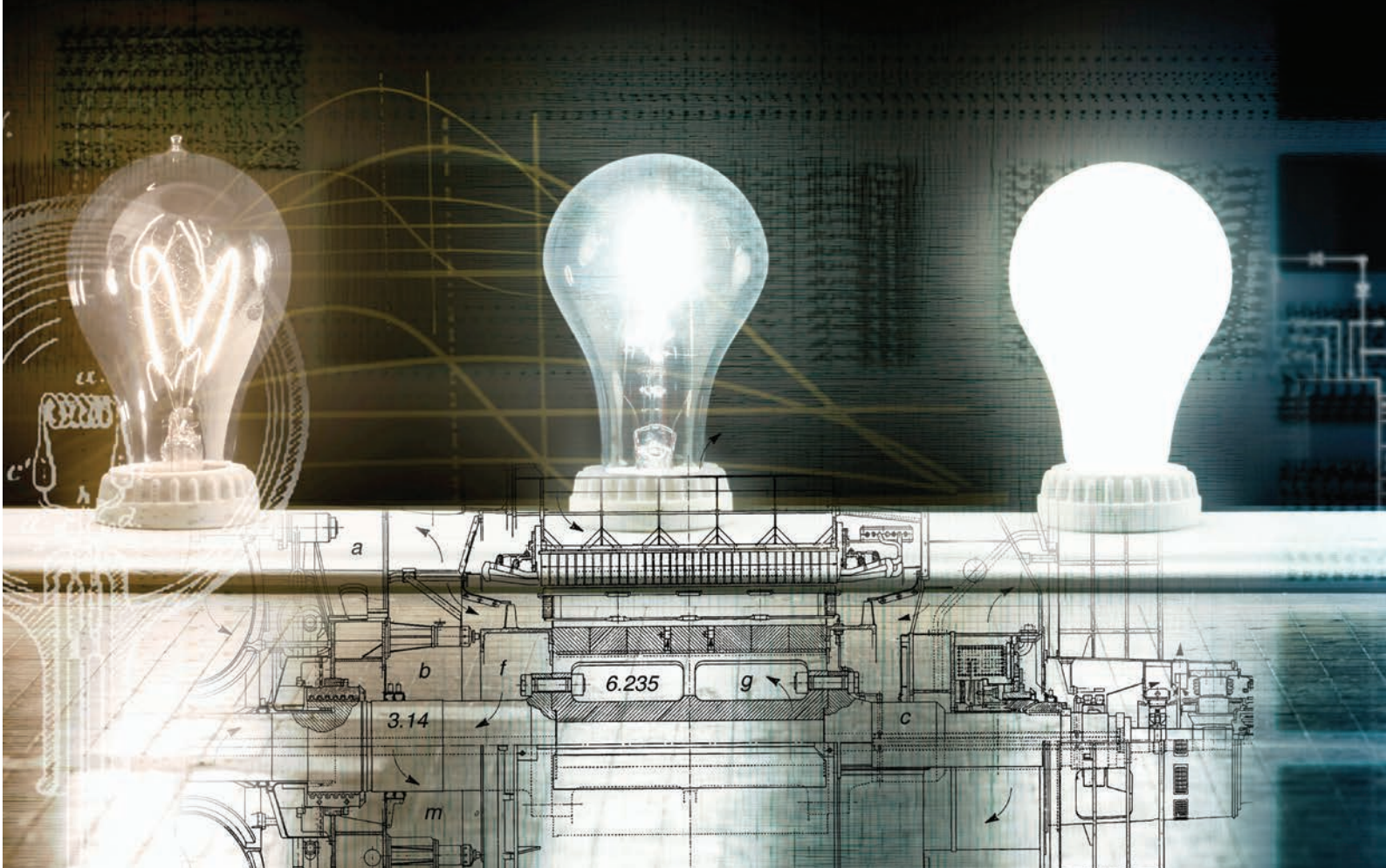


This catalogue shows the core range of Eaton Emergency Lighting (Self-contained) portfolio which covers a wide range of applications.

Besides this portfolio, Eaton works very closely with its customers in order to develop and manufacture products which meet Industries' very specific requirements.

For example, Eaton provides luminaires with minimal glare effect for train conductor specifically developed for tunnel applications, ATEX (explosion proof) products for harsh environments.

Several Eaton luminaire's specifications also meet food industry specific health and safety requirements (HACCP or IFS), like iP-65+, Atlantic LED or 46811 LED CGLine+.



From Static to Adaptive Evacuation, Eaton innovates to light the way to safety

Static Emergency Lighting

Eaton's quality above standards

Increased Affordance (IA)

For better visibility of escape routes
Learn more about IA on pages 8

Adaptive Evacuation (AE)

To show the safest way to exit
Learn more about AE on pages 10

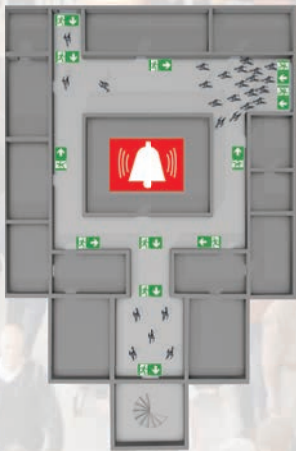
Innovation is key at Eaton. We own more than **30 patents** linked to emergency lighting.

- 2000: First to market with LED Emergency Lighting luminaires
- 2001: First adressable central power system
- 2003: First eco-friendly range of luminaires
- 2005: First monitoring panel for self-contained luminaires with web server integrated
- 2012: First graphic webserver
- 2015: Launch of CrystalWay luminaires, the new aesthetic reference
- 2017: Increased Affordance capability for better visibility of escape routes
- 2018: Adaptive Evacuation systems for safer evacuation



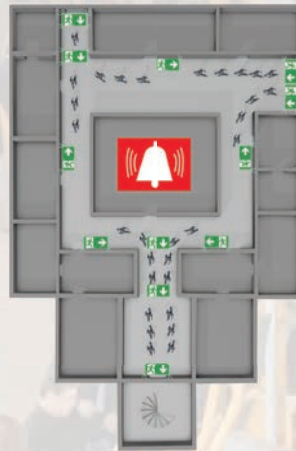
How to improve the recognition of escape routes for faster evacuation ?

Traditional evacuation



- Occupants may go to the exit they came in
- Crowding and congestion may slow evacuation
- Occupants may spend time trying to locate an exit
- Evacuation takes longer

Increased Affordance



- Provides a catalyst for less congested exit routes
- Aids people with hearing difficulties
- Ensures that users understand the importance of the emergency
- Compliant with emergency lighting standards

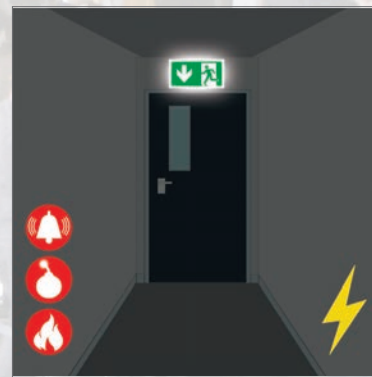
How does it work ?



In mains operation, **uniform illumination** of the pictogram facilitates the identification of emergency exits.



An evacuation order (fire, danger or climatic risk) triggers the **Increased Affordance** function which enables better recognition of the escape route through blinking or pulsing.



In case of power failure, the pictogram is perfectly legible **without risk of glare** for users.

For more information, refer to page 174

Adaptive Evacuation

Eaton innovates for safer evacuation

Improving escape route guidance

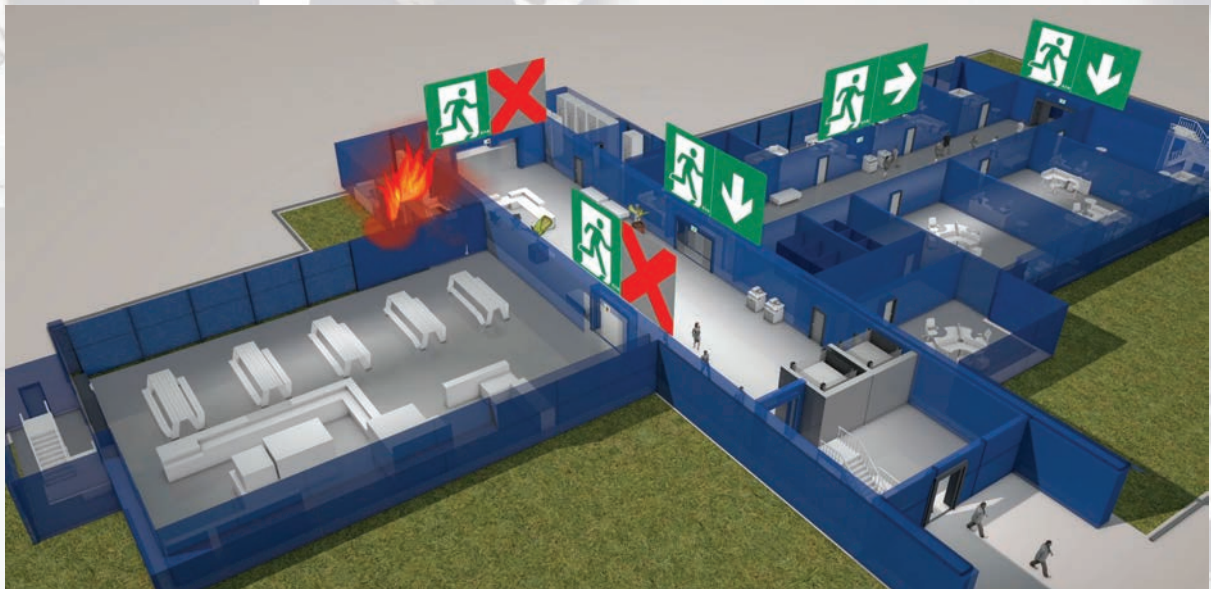
An increasingly urbanised and complex environment, combined with a rising diversity of safety threats, means the owners and managers of buildings need to re-evaluate the way they protect people and property as well as ensure business continuity that may be at risk in an emergency.

Building evacuation can prove increasingly challenging when a proportion of visitors are not familiar with layout and procedures, and particularly if they are in large, highly-populated, high-risk or complex premises such as railway stations, shopping centres, airports, stadia, government buildings or leisure facilities.

Research into crowd behaviour and advances in scenario-modelling technology have highlighted the need for evacuation strategies that are more adaptable to differing circumstances.

Building upon decades of expertise in the delivery of life safety systems, Eaton has developed a highly innovative Adaptive Evacuation System.

Adaptive signage enables building owners to direct people out of the building in the safest way as and when the nature of the threat changes. The danger posed by fires, acts of terrorism and natural disasters will change as the event unfolds; Eaton's Adaptive Evacuation signage allows building occupants to be redirected to the safest exit route available should such an event occur.



For more information, refer to page 180

Meet the innovative self-contained Matrix CGLine+ luminaire



Matrix CGLine+ can adjust the direction of the arrow it displays, with the aim of improving the speed and safety of emergency evacuations in commercial buildings.

The self-contained Matrix escape sign luminaire displays an arrow that can point in any one of four different directions. The additional ability to scroll the arrow from one side to the other increases its visual impact among the general population and helps to alert people with hearing impairments. It can also display a red cross to indicate that a particular exit route has become closed, blocked or dangerous. Making it a powerful tool when containment measures are the safest option. In such an emergency, it will not only lead occupants to but also keep them in a safe place.

How is the Matrix CGLine+ luminaire activated to adapt to the safest escape route?

In normal mode, the Matrix exit sign works exactly as a standard exit sign luminaire would, with all the necessary regulatory compliance.

In adaptive mode, they are able to indicate one of a number of different directions, based on a series of pre-programmed scenarios. This functionality can be activated manually from a central control room, based on CCTV footage, for example, or can be triggered automatically by specific local devices such as smoke alarms or panic buttons.

How is the self-contained adaptive-enabled CGLine+ system installed and maintained?

Ease of installation is always a key consideration during the design process and Eaton's self contained Adaptive Evacuation System is no exception. The adaptive-enabled CGLine+ is installed in the same way as its predecessor, but with additional inputs and back-up power supply. Operation of the system is reliant upon sophisticated cause-and-effect programming which should only be carried out by an appropriately trained specialist.

Ongoing maintenance of the system is aided by an automatic testing function, which runs continuous checks and issues alerts when any fault is identified. This enables remedial work to commence immediately and overcomes the time, cost and inconvenience associated with carrying out manual checks on a regular basis.

